Sets 3 — Probability from Venn diagrams · Conditional probability

Marks in brackets. ★ = harder question. $P(A) = \dfrac{n(A)}{n(\mathscr{E})}$; $P(A \mid B)$ = probability of A *given* B. Give probabilities as fractions in simplest form.

1 In a class of 30 students, 21 passed a Physics test (P) and 24 passed a Chemistry test (C). 2 students failed both. [5]

a Draw a Venn diagram.

b A student is chosen at random. Find **i** $P(\mathrm{P})$ **ii** $P(\mathrm{P} \cap \mathrm{C})$ **iii** $P(\mathrm{P} \cup \mathrm{C})$ **iv** $P(\mathrm{P}' \cap \mathrm{C})$.

2 A café records what 150 customers buy one morning: 120 buy a coffee, 70 buy a cake, and every customer buys at least one of the two. A customer is picked at random. Find the probability that they bought [5]

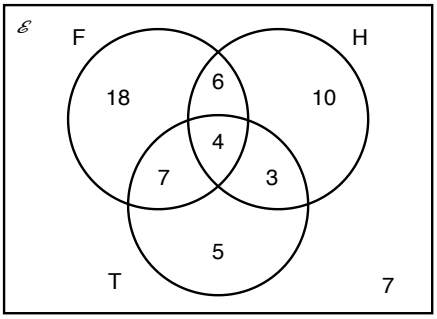
a a coffee and a cake **b** a coffee or a cake but not both.

3 In a survey, 80% of students own a bike (B), 30% own a skateboard (S) and 15% own neither. A student is chosen at random. Find the probability that the student [5]

a owns a bike or a skateboard but not both

b owns a skateboard, given that they own a bike.

4 The Venn diagram shows which of football (F), hockey (H) and tennis (T) the students in a year group play. [5]



a How many students are in the year group?

b A student is chosen at random. Find **i** $P(\mathrm{H})$ **ii** $P(\mathrm{F} \cap \mathrm{H} \cap \mathrm{T})$ **iii** $P(\mathrm{T} \mid \mathrm{H})$.

5 A music club has 40 members, each of whom plays at least one of piano (P), guitar (G) and drums (D). 24 play piano, 20 play guitar and 15 play drums; 9 play piano and guitar, 7 play guitar and drums, 8 play piano and drums. A member is chosen at random. Find [5]

a $P(\mathrm{P} \cap \mathrm{G} \cap \mathrm{D})$ **b** $P(\mathrm{D} \mid \mathrm{G})$.

Total: 25 marks

Check: $P(A \cup B)$ counts the overlap once · "but not both" = the two outer parts · $P(A \mid B) = \dfrac{n(A \cap B)}{n(B)}$ — the denominator is the *given* set, not the whole universal set · percentages work like counts out of 100 · simplify every fraction.